



RE26 – Hydropower

The RE26 Hydropower system provides students with a self-contained modular system, covering all aspects of Hydropower generation.

The Armfield advanced renewable energies range is a specialised learning system for the technical education in universities and vocational training centres.

The kits cover Photovoltaic and Wind Energy generation, Fuel Cell and Battery Technology, Thermal Energy and the construction of a controllable Smart Grid on a laboratory scale.

PRODUCT CODE: RE26

Category: RE Series - Renewable Energy

DESCRIPTION

The RE26 Hydropower system provides students with a self-contained modular system, covering all aspects of Hydropower generation.

Included in the system are various components that allow students to not only experiment how key principles of hydrostatics and fluid dynamics affect hydropower systems but how different turbine types affect the overall system outputs.

The self-contained RE26 provides comprehensive step by step manual taking students through every step, challenging them both practically and theoretically utilising key engineering mathematical principles, to support practical outcomes.

TECHNICAL SPECIFICATIONS

Turbine Casing: 2 inflows (upper/lower), 1 outflow; compatible with Pelton, Crossflow & Water Wheel

Turbine Set: Includes Pelton, Crossflow, and Water Wheel turbines

Manometer Sets: 2 bar for upper inflow, 4 bar for lower inflow

Intake Connector: Connects both inflows; valve-controlled

Flow Set (4 / 8 / 12 mm): 1m hose with 4 / 8 / 12 mm ID, valve, and O-ring marker

Aluminium Case: Storage and transport case

Induction Generator: Magnet disc-driven, generates DC using coils, diodes, and capacitors

AV-Module: Measures voltage (0–12V) & current (0–2A); 192×192 px display; battery-powered; optional USB/Wireless

Light Bulb Module: Plug-in micro bulb for basic load simulation

FEATURES & BENEFITS

- Experimentation system for solar thermal energy conversion
- Quantitative experiments for different collector systems

- Flexible and location-independent usage
- Laboratory scale
- Modular design
- Supplied in a self-contained aluminium case
- Includes in-depth manual and predefined experiments

Demonstration Capabilities

- Volume flow rate, flow velocity, and power as a function of the height of fall
- Volume flow rate, flow velocity, and power in relation to pipe cross-section
- Functional comparison of Pelton turbine, crossflow turbine, and waterwheel
- Performance comparison of the Pelton turbine, crossflow turbine, and waterwheel based on volume flow and pressure

Other Products in Range

- RE10: Advanced Photovoltaic Energy
- RE12: Advanced Wind Energy
- RE14: Advanced Fuel Cell Technology
- RE16: Advanced Thermal Energy
- RE18: Advanced Smart Grid Technology

Related curriculums

- Renewable Energies
- Electrical Engineering
- Automotive Engineering

Requirements

Electrical supply: 110-230V AC 50-60Hz

Level and stable work surface

Shipping Specifications

PACKED AND CRATED SHIPPING SPECIFICATIONS

Volume: 0.038m³

Gross weight: 6.5Kg

Dimensions

Length: 0.640m

Width: 0.165m

Height: 0.370m



Order Codes

RE26: Hydropower

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ADVANCED
PHOTOVOLTAIC
ENERGY**

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